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Digitizing SMEs in the EU: A scalable model for retrofitting machinery to Industry 4.0

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ABSTRACT

Small and Medium-sized Enterprises (SMEs) have faced significant economic challenges since the COVID-19 pandemic, compounded by high inflation, rising energy costs, and the cessation of government financial support. War-related disruptions in Europe have further exacerbated these issues. As the backbone of the European economy, SMEs must embrace Industry 4.0 to remain competitive. However, many lack the resources or expertise to adopt smart technologies. This article proposes a cost-effective, scalable model to guide SMEs in transitioning towards Industry 4.0 by retrofitting existing machinery. By assessing machinery readiness and identifying critical areas for digitization, the model enables targeted investments that optimize production and reduce costs. A gradual, tailored approach ensures alignment with SMEs' financial and technical capacities, fostering modernization and long-term growth. The successful implementation of the AFB production system highlights the feasibility of retrofitting existing systems to meet modern manufacturing demands.

1. Introduction

Small and Medium-sized Enterprises (SMEs) in Europe and globally have confronted unparalleled economic uncertainty and upheaval since the onset of the Covid-19 pandemic in early 2020. After the initial shockwaves of the pandemic subsiding, throughout 2021, SMEs encountered challenges in recruitment to accommodate an unexpectedly robust resurgence in demand, alongside coping with sharp and rapid escalations in the costs of numerous inputs. Although 2022 witnessed sustained economic recuperation, novel barriers that compromise the sustainability of entrepreneurial endeavors, particularly SMEs, surfaced. Elevated inflation rates, escalated energy and raw material expenditures, and the cessation of the pandemic-era fiscal assistance from governments exposed many SMEs to significant new risks. Furthermore, SMEs within the European Union (EU-27) were directly impacted by war conflicts, albeit in a circumscribed context through sanctions, export limitations, and disrupted supply chains, as well as indirectly affected by broader war-related developments such as soaring energy expenses, which imperil their expansion prospects.[1]

The statistics presented in this article are based on the Annual Report on European SMEs 2022/2023 of the European Commission (Di Bella et al., 2023). According to statistical data, SMEs comprise the most significant proportion of economically active entities within the European Union and can be considered the backbone of its economy. Many are equipped with state-of-the-art technology that meets the Industry 4.0 standard, yet they also possess technical equipment that does not meet one or more smart factory requirements. There are several barriers to their replacement. The main ones include the economic aspect of replacement with modern equipment, the satisfactory technical condition of the existing equipment, and the lack of knowledge of the possibilities of analyzing operational parameters. Transforming SMEs to the standards of Industry 4.0 can be challenging however is strategically necessary. Creating a universal platform for upgrading existing machinery and equipment will make such a process more manageable. First, SMEs must comprehensively analyze their current machinery and systems to assess their readiness for integration with

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[1] Di Bella, L., Katsinis, A., Lagüera-González, J., Odenthal, L., Hell, M., Lozar, B. Annual Report on European SMEs 2022/2023, Publications Office of the European Union, Luxemburg, 2023, doi:10.2760/028705, JRC134336. Licensed under a Creative Commons Attribution 4.0 International (CC-BY 4.0) license (http://creativecommons.org/licenses/by/4.0)

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smart technologies. This analysis serves as a starting point for ensuring that subsequent upgrades are implementable and beneficial.

Subsequently, the potential for upgrading the manufacturing process must be identified, focusing on sensor integration, increased connectivity, data processing, and storage capabilities. Such digitization aims to optimize production, increase efficiency, and reduce operating costs. By identifying the most influential potential (critical points) for digitization, SMEs can prioritize investments where digitization will benefit most. Thus, a SME does not need to digitize data collection across the entire production process. As one of the critical factors in SMEs' decision to invest in further development is the financial impact of this investment, a thorough cost–benefit analysis must follow, weighing the expenditure on modernization against the anticipated benefits. One of the critical factors in deciding on a SME's investment for further development is its financial impact. This must be followed by a thorough cost–benefit analysis that weighs the expenditure on upgrading against the anticipated benefits. From a long-term financial perspective, SMEs need to systematically upgrade their facilities to meet the requirements of Industry 4.0 and thrive in the era of smart factories.

The approach proposed in this study is designed to be flexible, scalable, and responsive to challenges and opportunities. However, the implementation of new technology systems is costly and technically challenging. This can be a barrier, especially for small and medium-sized enterprises with limited resources. The proposed approach is particularly suitable for SMEs that prefer a more cautious and gradual approach as they are more susceptible to the negative impacts of external circumstances such as market changes, new regulatory measures, or natural disasters. Therefore, it is crucial to have clear objectives and expected outcomes for Industry 4.0 implementation. If companies do not know or are unsure of what they expect from technology implementation, they may encounter several problems. This article analyzes the SME needs area and proposes a model for upgrading existing equipment to suit the Industry 4.0 concept and deployment in a smart factory environment. The proposed solution is characterized by a way of implementing Industry 4.0 technologies that are financially and technically affordable but at the same time adheres to all the principles of these principles and processes and introduces a proposal for a universal system for the digitalization of equipment in the production process, which aims to support SMEs in the process of digitalization and transformation of the enterprise into a Smart Factory following the Industry 4.0 standards.

2. Materials and methods

At the center of motivation for writing this article lies the recognition of SMEs as the backbone of the European economy, comprising over 99 percent of all businesses and employing a substantial portion of the workforce. Despite their economic significance, SMEs encounter numerous challenges in navigating a complex technological innovation landscape. Limited financial resources, a shortage of skilled labor, and the perceived risks associated with digitalization present formidable obstacles to SMEs seeking to embrace Industry 4.0. Integrating Industry 4.0 technologies holds immense promise for revolutionizing manufacturing processes, driving efficiency gains, and fostering innovation across industries. However, larger corporations have increasingly embraced these transformative technologies, SMEs often face significant barriers to adoption. They are located across many industrial areas although are united by the need to adapt their existing equipment and capabilities to match the smart factory requirements defined in the context of Industry 4.0.

Table 1
Statistics on EU enterprises.

Size	Number of enterprises		Number of employees		Value added	
	Number	Share	Number	Share	Billion €	Share
Micro	22 744 173	93,5%	38 790 351	29,4%	1419,4	18,6%
Small	1 332 200	5,5%	25 602 334	19,4%	1259,8	16,5%
Medium	204 786	0,8%	20 493 722	15,5%	1266,5	16,6%
SMEs	24 281 159	99,8%	84 886 407	64,4%	3945,8	51,8%
Large	43 112	0,2%	46 918 978	35,6%	3673,8	48,2%
Total	24 324 271	100,0%	131805385	100,0%	7619,6	100,0%

2.1. Statistical overview of SMEs in the EU

In the European Union, more than 24 million businesses were active in 2022, which can be characterized as SMEs, as shown in Table 1. These enterprises employed almost 85 million people and generated added value of nearly €4000 trillion.

Generally, an enterprise is considered a SME if it employs fewer than 250 employees and has an annual turnover of less than EUR 50 million or a balance sheet total of less than EUR 43 million. SMEs are further subdivided into the following categories:

- Micro enterprises, which have up to 10 employees,
- Small enterprises, which have between 10 and 49 employees,
- Medium enterprises, which have between 50 and 250 employees.

SMEs can be further classified according to the level of knowledge required and the technological level into the following categories:

- Low knowledge-intensive industries (LKII),
- Knowledge-intensive industries (KII),
- Low-tech industries (LTI),
- Medium-tech industries (MTI),
- High-tech industries (HTI).

The distribution of small and medium-sized enterprises (SMEs) across economic sectors, as illustrated in Fig. 1, provides valuable insights into the landscape of business activities within the EU. Notably, the data reveals a substantial presence of SMEs operating in sectors characterized by varying degrees of technological sophistication and knowledge intensity. Specifically, the analysis highlights the prevalence of SMEs in the KII, MTI, and HTI sectors, which collectively constitute more than 35 percent of the total SME population. KII encompasses diverse sectors where intellectual capital, research and development, and innovation are central to driving competitiveness and value creation. Within this sector, SMEs engaged in activities such as software development, consultancy services, creative industries, and information technology, leveraging their specialized knowledge and expertise to deliver high-value solutions to clients and customers. In parallel, the MTI represents a vital segment of the SME landscape, comprising businesses involved in manufacturing processes and industrial services requiring moderate technological sophistication. These industries encompass sectors such as machinery and equipment manufacturing, automotive components, and advanced materials, in which SMEs contribute to supply chain resilience, technological diffusion, and innovation diffusion within the broader industrial ecosystem. The HTI sector encompasses SMEs at the forefront of technological innovation and scientific advancement, driving breakthroughs in fields such as biotechnology, pharmaceuticals, aerospace, and electronics. SMEs within this sector are characterized by their focus on research-intensive activities, product development, and the commercialization of cutting-edge technologies, contributing to economic growth, job creation, and global competitiveness.

By highlighting the prevalence of SMEs across these knowledge-intensive and technology-driven sectors, Fig. 1 underscores their critical role in driving innovation, fostering economic growth, and enhancing the EU's competitiveness in the global marketplace. Moreover,

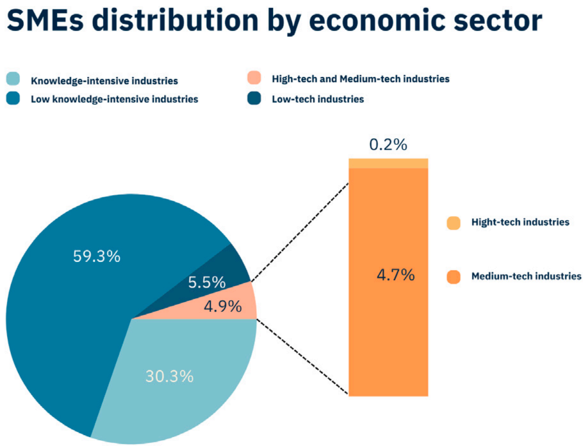

Fig. 1. SMEs distribution by economic sector.

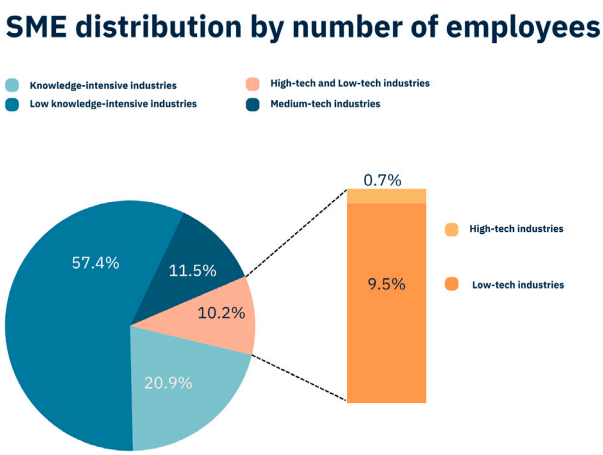

Fig. 2. SMEs distribution by number of employees.

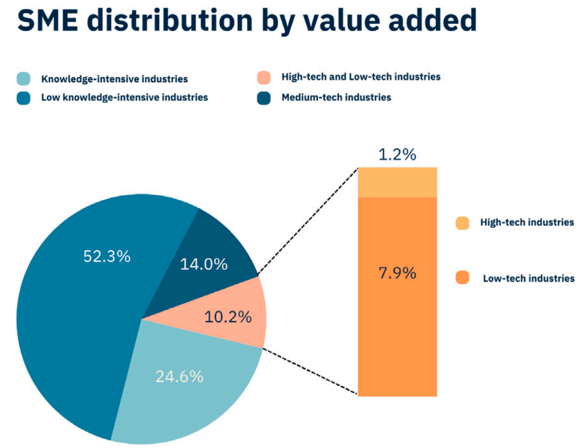

Fig. 3. SMEs distribution by number of employees.

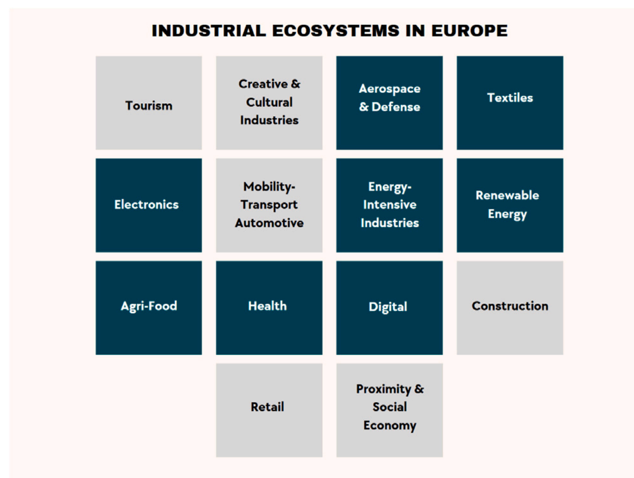

Fig. 4. Industrial ecosystems in EU.

it emphasizes the importance of targeted policies and initiatives to support SMEs in accessing finance, developing digital skills, and harnessing technological opportunities to unlock their full potential and to contribute to sustainable and inclusive economic development across Europe.

The distribution of SMEs by the number of employees is shown in Fig. 2. Companies in the KII, MTI, and HTI areas comprise more than 31 percent of the SME workforce. This highlights the need for targeted policies and initiatives to support SMEs in attracting and retaining talent, fostering a culture of innovation, and enhancing their competitiveness in the global marketplace. Strategic investments in education, training, and lifelong learning, policymakers can ensure that SMEs thrive and contribute to sustainable and inclusive economic growth across Europe.

The breakdown of SMEs by value added is shown in Fig. 3. Businesses in the KII, MTI, and HTI areas account for almost 34 percent of the total added SME value. By highlighting the substantial contribution of SMEs in knowledge-intensive and technology-driven sectors to total value added, Fig. 3 underscores the importance of fostering an enabling environment that supports innovation, entrepreneurship, and digitalization. According to [1], SMEs in the EU economy generate 51.8 percent of value-added, employ 64.4 percent of employees, and represent 99.8 percent of the total number of enterprises. SMEs play a crucial role in every industrial ecosystem cluster, as they constitute more than 99 percent of the enterprises in every ecosystem and can be considered the backbone of the EU economy. In March 2020, the European Commission noted, that "Europe also needs to look closely at industrial ecosystems' opportunities and challenges. These ecosystems encompass all players operating in a value chain: from the smallest start-ups to the largest companies, from academia to research, service providers to suppliers." The critical industrial ecosystems in the EU are displayed in Fig. 4 along with their main focus areas. According to European Union statistics, it shows the fourteen categories of industrial ecosystems in the European Union. In this particular case, blue tiles highlight the industrial ecosystems that will be dealt with in more detail in the manuscript. The gray tiles are beyond the scope of this study.

Currently, there are relevant EU competitiveness plans for standardization and digitalization. The "Rolling Plan for ICT Standardisation" (European Comission) provides a unique bridge between EU policies and standardization activities concerning information and communication technologies (ICT). This helps to increase the convergence of standards makers' efforts towards achieving EU policy goals. This document results from an annual dialogue involving a wide range of interested parties as represented by the European multi-stakeholder platform on ICT standardization (MSP). The Rolling Plan focuses on actions that can support EU policies and does not claim to be as complete as the work programs of the different standardization bodies.

The "Digitalisation in Europe" (European Commission, 2023) provides a comprehensive manual on digital transformation. It deals with people's digital skills, technology uptake in businesses, and e-government. Technology uptake in businesses aims to integrate digital technologies into all areas of a business, enabling companies to improve

Table 2
A structured comparison of strategic frameworks for Industry 4.0 adoption in SMEs.

Aspect	Shoestring philosophy	Retrofitting approach	Maturity model-based	RAMI 4.0 framework
Primary Focus	Low-cost digital solutions for specific needs	Physical machinery upgrading with sensors at automation levels	Assessment-led implementation based on capabilities	Standardized reference architecture
Implementation	Bottom-up, targeted interventions (McFarlane et al., 2019)	Three-stage BPMN workflow (Gašpar et al., 2024)	Assessment, gap analysis, roadmap (Schuh et al., 2020)	Layer-based architecture integration (Hankel and Rexroth, 2015)
Cost Model	Extremely low investment with incremental additions	Moderate investment with immediate gains	Variable investment based on maturity	Substantial investment across layers
Integration	Standalone solutions with minimal disruption	Direct enhancement without modifying control	Systematic integration based on maturity	Standardized integration across layers
Data Use	Basic collection for specific insights	Focused collection at critical points	Comprehensive architecture by maturity	Unified models across hierarchy
Key Strength	Minimal investment with tangible benefits	Practical approach for bottlenecks	Systematic framework for implementation	Comprehensive interoperability
Limitation	Limited scope, potential fragmentation	May not address enterprise transformation	Often theoretical with lengthy assessment	Complex requiring significant resources

their products and services, and gain competitiveness, for example, by shifting their sales online. The EU has set two primary goals for the digital transformation of businesses by 2030: more than 90 percent of SMEs should reach at least a basic level of digital intensity, and 75 percent of EU companies should use cloud computing services, perform big data analysis, or use artificial intelligence.

2.2. Different approaches to enhancing SMEs business to implementing Industry 4.0

In recent years, big business have been and is intently implementing the core pillars of the Industry 4.0 concept. While large enterprises have demonstrated significant progress in incorporating Industry 4.0 technologies into production processes, small and medium-sized enterprises (SMEs) face particular challenges, including resource constraints, limited digital competence of employees, and operational vulnerabilities during transition periods. Implementing Industry 4.0 in SMEs therefore requires agile tailored approaches that take these constraints into account while maximizing the potential benefits. A comparative analysis is shown in Table 2. It presents four prominent methodological frameworks for Industry 4.0 implementation in SMEs found in the literature: the Shoestring philosophy, the Retrofitting approach, the Maturity Model-based implementation, and the Reference Architecture Model for Industry 4.0 (RAMI 4.0) framework.

The different approaches represent different philosophical foundations, methodological approaches and resource implications that significantly affect their application domain. This comparative analysis contributes to the scientific overview of Industry 4.0 implementation methodologies for SMEs by comparing different approaches along critical dimensions. By situating our modernization strategy within the broader environment of implementation frameworks. This comparison better contextualizes the distinctive characteristics and advantages of each approach, while also contextualizing their applicability in different organizational contexts. The modernization methodology, with its pragmatic focus on improving existing physical infrastructure through targeted digital augmentation, emerges as a particularly viable path for manufacturing SMEs with significant capital investments in legacy facilities. Nevertheless, comparative framework not only advances the theoretical understanding of digital transformation pathways in resource-constrained environments, but also provides practitioners with evidence-based guidance for strategic decision-making aligned with organizational capabilities and goals. As mentioned earlier, the typology we have presented deals with four distinct methodological frameworks. In this case, it is important to note that various complementary implementation elements can be integrated within these primary approaches. Platform-based technology systems, consortia collaboration Qin and Yang (2008), service-oriented business models Adrodegari et al. (2018), targeted technology deployment Horváth and Szabó (2019) and business partnerships for knowledge transfer Harris (2009) are tactical elements that can increase the efficiency of implementation and consequently achieve synergies in the outcome.

The literature on Industry 4.0 implementation for SMEs exhibits diverse methodological approaches. These diverse perspectives on Industry 4.0 adoption in SMEs offer complementary insights. Müller et al. (2024) provide a global understanding of barriers through their multi-country study of 15 SMEs, identifying 12 key barriers within the Technology-Organization-Environment framework. Semeraro et al. (2023) contribute a quantitative assessment methodology using fuzzy logic to measure implementation maturity across nine technologies and six principles. These approaches offer valuable insights into challenges and readiness assessment. Building upon existing research foundations, this retrofitting model addresses the practical implementation gap by providing a process-oriented framework for machinery modernization that accommodates the resource constraints identified in previous studies. Table 3 presents a comparative analysis of three distinct methodological approaches to Industry 4.0 implementation in SMEs, highlighting how these perspectives address different aspects of the digital transformation journey for smaller enterprises with limited resources.

2.3. Technological overview of practices for upgrading existing equipment

Several studies have been published on the issue of existing equipment technology upgrades from various perspectives, deployment options, and country-specific requirements. In Nowacki et al. (2021), the authors state that to remain competitive in the future, today's companies must change their production strategy towards Industry 4.0 and question, therefore, such a change of strategy managed in a structured way in the brownfield. Another example is in Kröll and Cseh (2023), where the authors dealt with sustainable production systems, representing an important factor contributing to reducing bluehouse gas emissions and resource efficiency for the industrial sector. The claim that digitalizing legacy factory equipment using a digital retrofit constitutes a significant building block for this goal. Carlo et al. (2021) and Ilari et al. (2021) presented a framework for retrofitting a process plant applied to a two-phase mixing plant, providing a guide starting from the plant study and ending with the I4.0 paradigm implementation. This approach allows for improved plant safety and maintainability. Another example (Deshpande et al., 2023) is plastic and composite manufacturing to overcome the complexities associated with these materials. The authors claim that retrofitting legacy

Table 3
Key aspects in methodological approaches to industry 4.0 integration for SMEs.

Feature	Müller et al. (2024) (Müller et al., 2024)	Semeraro et al. (2023) (Semeraro et al., 2023)	Gašpar et al. (2024) (Gašpar et al., 2024)	Key differences
Primary Focus	Barriers and enablers for Industry 4.0 integration in SMEs across global context	Maturity model for evaluating impact of Industry 4.0 technologies and principles	Scalable model for retrofitting existing machinery to Industry 4.0 standards	Focus: • identifying barriers (Müller) • measuring maturity (Semeraro) • practical implementation (Gašpar)
Research Approach	Multiple case study	Maturity model development and validation in UAE SME	BPMN model development and real-world implementation	Focus: • international comparative analysis (Müller) • quantitative measurement (Semeraro) • process modeling and physical implementation (Gašpar)
Conceptual Framework	Technology-Organization-Environment (TOE) framework	Fuzzy logic-based maturity model	Three-stage BPMN model	Different focuses on theoretical bases: • organizational factors (Müller) • maturity levels (Semeraro) • process workflow (Gašpar)
Technology Focus	Broad range of I4.0 technologies, emphasis on integration	Nine technologies correlated with six design principles	Focus on physical retrofitting at field, control, and supervisory levels	Focus: • comprehensive (Müller) • principle-oriented (Semeraro) • hardware-focused (Gašpar)
Implementation Level	Theoretical with empirical validation	Measurement-oriented with case application	Practical implementation with real-world testing	Progresses: • from conceptual (Müller) • to measurement (Semeraro) • to practical (Gašpar)
Evaluation Method	Cross-case analysis of interview data	Fuzzy logic weighted assessment	Real-world implementation and performance metrics	Different validation approaches: • qualitative (Müller) • quantitative (Semeraro) • performance-based (Gašpar)
Geographic Scope	Global	Single country	EU-focused	Wide to narrow geographic focus
Decision Support	Guidance on barrier identification and enabler selection	Decision support for technology investment prioritization	Detailed workflow for retrofit implementation	Decision support levels: • strategic (Müller) • tactical (Semeraro) • operational (Gašpar)
Time Horizon	Long-term strategic planning	Medium-term technology roadmap	Immediate practical implementation	Different planning horizons

manufacturing systems with modern sensing and control systems is emerging as a cost-effective approach, as it circumvents the substantial investments needed to replace legacy equipment with expensive modern systems to enhance productivity. In Bazen et al. (2022), the authors presented a possible implementation of the Industry 4.0 standard in the Netherlands, based on the development of a smart factory blueprint for producing energetic retrofitting packages. In Alqoud et al. (2022a) and Alqoud et al. (2022b), the authors addressed the issue of retrofitting existing devices using available sensors and IoT solutions and identified three ways to implement such upgrades. From the above studies, it can be concluded that, before starting any upgrade initiative, it is imperative to commence with a comprehensive assessment of the existing technological infrastructure within SME. This assessment extends beyond merely examining production equipment, encompassing an in-depth analysis of prevailing energy systems, information and communication infrastructure, and spatial and environmental considerations pertinent to the site. The evaluation process serves as the foundational step in gauging the current technological prowess of SME and provides invaluable insights into areas ready for enhancement and optimization. By scrutinizing every facet of the technological landscape, from machinery and utilities to digital infrastructure and environmental factors, stakeholders can better understand the existing capabilities and constraints that shape the enterprise's operational landscape. Moreover, it is imperative to translate a SME's vision for technological advancement into a meticulously curated catalog of requirements. This catalog serves as a blueprint, delineating the specific needs and aspirations of the enterprise in detail. From performance benchmarks and operational specifications to regulatory compliance and sustainability objectives, each requirement is precisely articulated to ensure alignment with the overarching strategic objectives of SME. The final implementation phase ensues with the functionality requirements firmly established, wherein the identified enhancements and optimizations are systematically realized. The implementation phase ensures the seamless integration of cutting-edge solutions tailored to meet the bespoke needs by leveraging a strategic blend of technological expertise, project management acumen, and stakeholder collaboration. In essence, the iterative process of analysis, requirement formulation, and implementation epitomizes a holistic approach to technological upgrade initiatives within SMEs. By adhering to this structured methodology, stakeholders can confidently navigate the complexities inherent in technological transformation, ensuring the successful realization of their strategic objectives and positioning SME for sustained growth and competitiveness in the digital age. For the purposes outlined in the preceding sections, the authors advocate the adoption of a scalable model tailored for retrofitting machinery to adhere to Industry 4.0 standards. This model is pivotal in furnishing the technology upgrade investigator with a structured workflow, meticulously guiding the process from its foundational elements to its culmination. By comprehensively delineating each step, the model ensures that the resulting transformation is unequivocally defined, executable, and subject to rigorous verification and validation processes.

2.4. Overview of existing technological upgrade solutions

This section undertakes a comprehensive analysis of existing cases and categorize them based on their level of digitization into distinct classifications. The first category, devoid of digitalization, encapsulates cases in which traditional methods and analog systems predominate, lacking any integration of digital technologies. These instances typically rely on manual processes, physical documentation, and conventional machinery devoid of digital interfaces. By contrast, the semi-digitalized category encompasses cases in which digital technologies have been partially integrated into existing systems. While certain aspects may exhibit digitization, such as basic automation or data collection, significant portions of the operations may still rely on traditional methods and analog processes. The final category, fully digitalized, represents cases in which digital technologies have been comprehensively integrated across all aspects of operations. These instances feature advanced automation, real-time data analytics, interconnected systems, and sophisticated digital interfaces, which facilitate streamlined processes, enhanced efficiency, and greater adaptability to changing demands. Commencing the technology upgrade at the lowest "field level" is a crucial starting point, as the technological apparatus here delineates the system's fundamental attributes, as such devices are directly involved in operating and controlling physical processes. Primarily encompassing an array of sensors, controls, actuators, drives, and analogous devices inherent to this technology level, the field level is the bedrock upon which subsequent advancements have been made. By initiating the upgrade process at this foundational tier, the inherent functionalities and capabilities of the system are fundamentally reshaped and augmented, paving the way for the subsequent layers of enhancement and integration. It not only improves current operations but also makes it easier to integrate advanced technologies. It is possible to find many articles supporting these considerations. In Keshav Kolla et al. (2022), the authors stated that retrofitting the existing machinery with sensors and building an Industrial Internet of Things (IIoT) is more beneficial than upgrading the equipment to newer machinery. They demonstrated this on a simplistic retrofitting of a drilling machine with no sensors or information-gathering capabilities. In Hemmelgarn et al. (2021a), the authors correctly stated that the sensors provide the necessary input variables for information processing and thus form the essential key component for acquiring the primary system and the environment. They suggested the use of 3D Mechatronic Integrated Devices (MID) and Additive manufacturing (AM) for small-batch sensor manufacturing. They stated that the combination of MIDs, and AM not only fulfills the requirements regarding costs and flexibility but also enables the creation of new, highly individual, and electrically functionalized products in general. Another article (Pietrangeli et al., 2023a) discussed smart retrofitting by integrating various technologies such as sensors and control systems. This study aims to distinguish between different strategies and find the correct definition of smart retrofitting. It highlights the relevance, benefits, and sustainability of Digital Twin solutions for smart retrofitting in the industrial sector. Technology upgrades envisage architecture for the seamless integration of straightforward solutions employing microcontrollers to assume a pivotal role in facilitating the intricate data acquisition process. This approach leverages the versatility and efficiency of microcontroller-based systems to streamline retrieval of essential data points. Primarily, the system orchestrates the acquisition process either at predetermined intervals or in response to detected changes in the sensor values, ensuring the timely capture of relevant information. This adaptive acquisition methodology will be tailored not only to suit the unique characteristics of the operational site, but also to align with the specific attributes of the sensors employed within the system. Several available examples of practical solutions to the underlying problems support this. For instance, in John and Vorbrocker (2020), regarding the question of retrofit, the authors assert that by integrating the measurement data from sensors into an IoT environment, more value can be derived from the data through advanced analytics using machine learning or deep learning. They claimed that the controller needs to be retrofitted with IoT gateways to enable IoT connectivity for sensors connected to Programmable Logical Controllers (PLCs), especially in the case of older-generation PLCs. Kopacek et al. (2020), developed tools to retrofit robotic components and legacy computer hardware with a combination of inexpensive hardware to help SMEs reduce costs and increase their productivity. The goal of their work was to evaluate the use of widely available open-source technologies such as the Linux Operating System and LinuxCNC, as well as the use of inexpensive hardware, such as Arduino, Raspberry PI and legacy hardware, for motor drivers to create controllers that can be used in a variety of different robot types in the virtual world and laying the groundwork to control a real robotic arm. An example of an affordable solution is the proposal (Di Carlo et al., 2021), where the authors used a relatively simple Arduino Mega 2560 microcontroller board as a versatile, reliable, easily programmable, and cheap solution. According to Nakakaze et al. (2022a, 2024), retrofitting is a sustainable means of equipping aged machines with low-cost sensors and microcontrollers to read and forward the machine data. In their work, the authors presented a concept and a prototype to retrofit industrial scenarios using lightweight web technologies on the edge. Rupprecht et al. (2021), designed retrofitting concepts that focus on access to PLC data to account for the unique, mission-critical role of PLCs. They aimed to fill this gap by deriving generally applicable retrofitting strategies for the data collection of legacy PLCs. Upon successfully retrieving data, a correctly designed protocol was enacted to systematically transfer the acquired information to a preordained repository. This repository, meticulously crafted to accommodate the diverse array of data generated by the system, serves as a centralized hub for aggregating and organizing the acquired data. Sophisticated communication protocols were employed to ensure optimal data integrity and transmission efficiency to facilitate a seamless data transfer process. In tandem with these efforts, Human–Machine Interface (HMI) technology will be leveraged to enhance user interactions and system monitoring capabilities. Finally, specialized programming tools were harnessed to orchestrate the intricate interplay between the microcontroller-based systems, communication protocols, and data repository, thereby ensuring the cohesive operation of the entire data acquisition framework. Several studies have addressed this issue. John and Vorbrocker (2020) stated that the cost of retrofitting older generation PLCs with IoT gateways and the extensive configuration and setup required to use sensor data in running environments can prevent industries and factories from utilizing the benefits of IoT. Their proposed architecture addresses these issues by describing an approach to capture sensor data from PLCs using the Modbus protocol, transforming the data into an IoT communication standard, and providing contextual information on data sources on demand to enable plug-and-play IoT connectivity. In García et al. (2022), the authors introduced a typical system architecture to allow modular communication between three tiers for collaborative maintenance in traditional manufacturing. These three conceptual tiers manage the collaborative digital retrofitting solution in a non-intrusive manner: the Edge tier, which interacts with the sensors, machines, and workers using retrofitting strategies; the Cloud tier, which provides SMEs with means to understand the maintenance needs; and the Business tier, which generates collaborative maintenance knowledge for workers and processes. Another exciting contribution (Reddy et al., 2023) proposed a framework for converting an older machine to an Industry 4.0-machine to provide real-time data such as feed rate, tool position, tool vibration, and cutting forces. The data collected from the sensors are transmitted by a server and stored in a database that can be used to create dashboards and digital twins. Communication over well-known architectures is discussed in Torres et al. (2021), where the authors say that communication can be achieved through the OPC UA or MQTT architecture, both of which align with the new trends for industrial communications. The sensor data will be used to feed production and manufacturing key performance indicators and for predictive maintenance. The approach presented in this study allows industries with legacy equipment to renew and adapt to new market trends, improve productivity rates, and reduce maintenance costs. The solutions implemented in Burresi et al. (2020) are described mainly regarding machinery interfacing by OPC for data acquisition, image processing for anomaly detection, and information visualization for human-machine interaction. This study stresses the need for cooperation between operators in the design process and the synergy between machinery-things-people inside the production plant. Concerning HMI and visualization, the authors in Kancharla et al. (2021) stated that two main assumptions can be made: retrofitting consists of adding a non-intrusive external system (a camera or vibration sensor) to the machine, and data must only be visualized on a singular device. This device is probably either a stationary monitor or AR headset. In another study (Atzeni et al., 2023), the authors used an anonymized dataset obtained from two medium-sized companies, leveraging a non-invasive and scalable data-collection procedure to demonstrate how machine learning (ML) techniques can enable SMEs to extract helpful information even in the short term, from a small variety of data types. This thoroughly designed architecture underscores a holistic approach to data acquisition, characterized by the fusion of cutting-edge technology, meticulous planning, and robust implementation strategies. This architecture promises to deliver a robust, scalable, and efficient data-acquisition framework that meets the diverse needs and challenges of modern industrial environments by harnessing the inherent capabilities of microcontrollers, sophisticated communication protocols, and specialized programming tools. At this point, we will provide examples of how to set up the architecture for the upgrade. Concerning upgraded architecture, the authors in their work (García et al., 2022) proposed designing the upgrade using four steps: goal definition, as-is analysis, upgrade integration, and user interface design. Another approach was presented in Polzer et al. (2023), based on the low-cost "Shoestring" philosophy of the Institute for Manufacturing, University of Cambridge. This study reports the retrofitting of a semi-automatic pipe-welding machine in a factory. A five-phase approach was adopted, from initial analysis, development, and procurement to commissioning and employee training. In Vuković et al. (2022), the authors presented a rapid, low-cost prototyping solution for manufacturing companies with legacy machinery intended to adopt the Industry 4.0 paradigm with a low-risk initial step. Legacy machines are retrofitted through the Industrial Internet of Things, making them both connectable and capable of providing data, thus enabling process monitoring. In Ralph et al. (2021), the authors focus on two main objectives: reducing the investment costs for smaller companies by using mainly open-source software (SW) and cost-effective but suitable hardware (HW), and the development of a resilient Cyber–Physical Production System (CPPS), which can be used to educate engineering students, and therefore, future production managers as well as other interested parties from the manufacturing industry segment in the topic of digitalization and associated technologies. In the following Table 4, we comprehensively examine and contrast diverse methodological approaches implemented by various authors in the extant literature of retrofitting implementation. This synthesized comparative analysis affords a more nuanced perspective on the current research landscape, thereby contextualizing our proposed framework within the established body of knowledge in this domain.

3. Proposal of a scalable model for machinery retrofit

In the process of drafting our proposal for a scalable model for machinery retrofitting, we have studied several publications. The authors in Pietrangeli et al. (2023a) discussed smart retrofitting, which involve the integration of various technologies such as sensors and automation systems. Their work aims to distinguish different strategies and find a correct definition of smart retrofitting, highlighting its relevance, benefits and sustainability in the industrial sector, focusing more on Digital Twin solutions for smart retrofitting. In Alimam et al. (2023), the authors look beyond Industry 4.0 and claim that Industry 5.0 is the Age of Augmentation and depict their approach to a digital triplet, which discloses the potency of their proposal of retrofitting a conventional drilling machine. The review of publications in Słowik and Sierocka (2023), according to the authors, confirmed the proposed division of the elements of retrofitting machines and devices into three components: sensory elements, clouds, and the Internet of Things (IoT) devices as communication elements, and artificial intelligence algorithms and machine learning techniques for drawing conclusions based on collected data. One solution from a technical perspective is Digital Retrofitting (Tantscher and Mayer, 2022), which focuses on the implementation of additional sensors and edge devices to digitally connect the machine and transforming it into a cyber–physical system. This study introduces a holistic procedure model for Digital Retrofitting, which is the result of an analysis and clustering of the existing process models. Maintenance strategies have evolved since the first industrial machines required repairs in the First Industrial Revolution claim by the authors in Sanchez-Londono et al. (2022). They state that these strategies have matured from simple corrective actions to more complex ones that support asset management and provide added value through the exploitation of information and computer technologies. Our authors' team decided to analyze in detail the retrofit options at the first three levels of the automation pyramid in terms of the necessary hardware extension, as shown in Fig. 5:

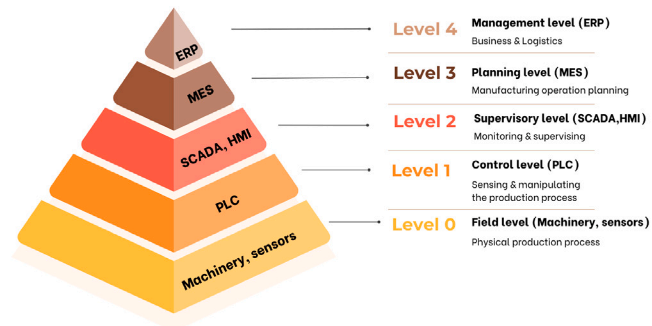

Fig. 5. Automation pyramid.

1. Field level: This layer represents the physical manufacturing process. At this level, we are concerned with devices and components that directly interact with the production environment, such as sensors, actuators, and other peripherals. Hardware enhancements at this layer may include upgrades to existing equipment to improve the accuracy and reliability of data collection, and the integration of new technologies to increase of the overall process efficiency.

2. Control level: This layer represents control and manipulation of the manufacturing process. Hardware focuses on control systems such as programmable logic controllers (PLCs), distributed control systems (DCSs), and other control units that process sensor data and issue commands to actuators. Retrofitting at this level may include installing more advanced control units that allow faster and more accurate information processing and better integration with modern information technologies.

3. Supervisory level: This layer represents monitoring and supervision of the manufacturing process. It focuses on systems that enable real-time data tracking and analysis as well as interfaces for operators and managers. Hardware extensions in this layer may include the implementation of advanced supervisory control and information systems (SCADA) that provide a comprehensive view of the production status and the integration of analytical tools for predictive maintenance and process optimization.

Each layer plays a crucial role in manufacturing automation and requires careful evaluation for retrofit planning and implementation. Our proposal includes a detailed examination of hardware requirements and expansion options to ensure optimal functionality and efficiency of the manufacturing system at every individual level.

Table 4
Comprehensive comparison of retrofitting approaches for industry 4.0.

Aspect	Gaspar et al. (2024) (Gašpar et al., 2024)	Rupprecht et al. (2021) (Rupprecht et al., 2021)	Di Carlo et al. (2021) (Di Carlo et al., 2021)	Pietrangeli et al. (2023) (Pietrangeli et al., 2023b)	Hemmelgarn et al. (2021) (Hemmelgarn et al., 2021b)	Nakakaze et al. (2022) (Nakakaze et al., 2022b)	Kopacek et al. (2021) (Kopacek et al., 2021)	John & Vorbrocker (2020) (John and Vorbrocker, 2020)
Primary Focus	Scalable retrofitting model for SMEs to transition to Industry 4.0	PLC retrofitting concepts for Industrie 4.0 scenarios	Process plant retrofitting for safety and maintenance	Smart retrofitting as sustainable solution	Evaluating AM technologies for MIDs in retrofit sensor systems	WebAssembly-based retrofitting for edge computing	Robot retrofitting using LinuxCNC with Arduino/RaspberryPI	IoT connectivity gateway for ModbusTCP sensors
Target Industry	SMEs across multiple industrial ecosystems	Manufacturing industry	Process industry	Manufacturing industry	Sensors for retrofit applications	Manufacturing with existing older PLCs	Industrial robotics	Process industry with ModbusTCP networks
Approach	BPMN model with 3-stage implementation process	Five technology-neutral retrofitting concepts	Four-phase approach with Digital Twin implementation	Differentiation between revamping, remanufacturing, and retrofitting	Evaluation of additive manufacturing technologies for MIDs	WebAssembly-based architecture for edge devices	Open source retrofitting of robotic controllers	Gateway architecture for ModbusTCP sensors
Key Technologies	Raspberry Pi, accelerometers, microphones, IoT	HAL (Hardware Abstraction Layer), PLC4X, stepgen	Digital Twin, Deep Learning, sensors	Digital Twin, AI, sensors, automation	MID Lacquer, MID Resin, CMID	WebAssembly, Docker, ESP32 microcontrollers	LinuxCNC, Arduino, Raspberry PI, V-REP simulator	MQTT, SenML, MongoDB, TICK stack
Implementation Scale	Cost-effective small-scale implementation	Three concepts validated in proof-of-concept	Plant-wide implementation	Various scales from individual machines to full systems	Component-level evaluation	Prototypical implementation on microcontrollers	From 2 DoF plotters to 6 DoF robots	Gateway connected to temperature/CO2 sensors
Benefits Emphasized	Reduced downtime, gradual approach for SMEs	Flexibility, interoperability, modular approach	Improved safety and maintenance	Sustainability (economic, social, environmental)	Cost-effective sensors for small batches	Near-native performance at edge, portability	Low-cost, virtual world simulation, training	Plug and play IoT connectivity, minimal setup
Cost Considerations	Explicitly addresses limited resources of SMEs	Focus on cost-effective solutions	Limited investment, reasonable implementation time	Cost-effectiveness as main advantage	Cost comparison of technologies	Low-cost hardware (€4 for ESP32)	Uses legacy motors, inexpensive hardware	Open-source components to reduce cost
Validation Method	Real implementation in AFB production system	Proof-of-concept with three concepts	Case study on two-phase mixing plant	Literature review	Technical evaluation of prototypes	Tested with temperature and CO2 sensors	Real implementation with pen plotter	Tested with temperature and CO2 sensors
Industry 4.0 Integration	Retrofitting at field, control, and supervisory levels	Integration with I4.0 concepts	Integration with DT and predictive maintenance	Integration of multiple I4.0 technologies	Component-level sensing capabilities	Integration with data lake setup	VREP simulator for virtual twin	Integration with data analytics
Limitations Addressed	Inability to modify existing PLC architecture	Data security, hardware constraints, complexity	Hardware adaptation challenges	Knowledge barriers and technical compatibility	Trade-offs between AM technologies	Limited resources on microcontrollers	Parallel port limitations, motor control	Communication protocols, semantic data context

3.1. BPMN model design procedure

In this section, we identify the participants and individual processes involved the retrofit execution. The participants were categorized into two groups. Table 5 outlines the participants representing SME customer, specifically: C. Management, Technology Officer, Operator, and Maintenance.

As SME customer, SME acts as end-users of digital technologies, seeking to modernize their internal operations through the adoption of Industry 4.0 solutions. These enterprises typically aim to enhance production efficiency, increase process transparency, and improve quality control by integrating technologies such as cyber–physical systems, industrial internet of things (IIoT) components, and data-driven monitoring tools. However, the implementation of such technologies is often constrained by limited financial and human resources, especially in smaller firms. In many cases, SMEs lack in-house expertise in advanced automation, software development, or systems integration, making them reliant on external providers. Furthermore, many SMEs operate with legacy equipment that was not originally designed for connectivity or digital integration, necessitating the development of cost-efficient, retrofit-compatible solutions. The decision to adopt Industry

Table 5
SME customer participants.

SME customer	
C.Management	Responsible for managing the company. Decides on technology update proposal.
Technology officer	Responsible technology used in production. Gathers ideas from stuff.
Operator	Responsible for using technology. Reports ideas and requirements to technology stuff.
Maintenance	Responsible for maintaining technology. Reports ideas and requirements to technology stuff.

Table 6
Retrofit supplier participants.

Retrofit supplier	
S.Management	Responsible for managing the company. Decides on price offers and planning.
Analyst	Responsible for assignment analysis and defining technology requirements.
Engineer	Responsible for selecting specific technical components based on analysis.
Integrator	Responsible for integrating technical components into existing technology.

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technologies is therefore typically guided by pragmatic considerations, with a strong emphasis on return on investment and minimal disruption to existing processes.

Table 6 defines the participants in the Retrofit Supplier. Specifically, these are S.Management, Analyst, Engineer, and Integrator. As a Retrofit supplier, SME participates in the Industry 4.0 ecosystem by providing critical components, specialized services, or complete technological solutions that support the digital transformation of industrial production. These firms may develop hardware (e.g. sensors, embedded devices), software (e.g. analytics platforms, user interfaces), or system integration services tailored to the needs of various industrial clients, including other SMEs and larger enterprises.

Operating in this role requires a high degree of technological competence and innovation capacity, often supported by collaboration with research institutions or participation in technology clusters. Such SMEs contribute significantly to the diffusion of Industry 4.0 principles by offering flexible, modular, and customizable solutions that address the diverse needs of the market. Their agility and proximity to clients often position them as valuable innovation partners, particularly in niche or rapidly evolving technological domains.

After defining the participants, we categorized the processes involved in the execution of the retrofit into the following three distinct stages. Each stage represents a critical phase in the overall workflow, ensuring a systematic and organized approach to retrofitting.

3.1.1. Stage 1 - SMEs identified needs

The first stage of model development involves a detailed definition of the SME needs, as shown in Fig. 6. These needs are determined based on a comprehensive analysis of the as-is state of the existing equipment as well as the expected requirements for new functionality. To define these needs, it is essential to consider the current technical requirements and the potential for future growth and expansion of the system. This means that when designing the model, we must anticipate technological and engineering advances that could affect the future needs of the business. Ensuring the flexibility and scalability of a facility is critical for its long-term sustainability and ability to adapt to changing conditions and technological innovations.

The verbal description of the BPMN diagram for Stage 1 is as follows: At the beginning, a task assignment for C.Management is set to retrofit the existing machinery (Retrofit task assignment). This is followed by a task to ensure the identification of the current machinery state (Check the current machinery state) by the Technology officer; then the Operator carries out the task to complete the operator's requirements (Identify the operator's requirements). Maintenance solves the task of completing the maintenance requirements (Identify maintenance requirements). Their outputs go into the next task (Processing of retrofit requirements) where the Technology officer processes them and adds the requirements based on the desired level of upgrade. This list represents retrofit requirements for the existing equipment that the C.Management finalizes (Retrofit requirements prepared).

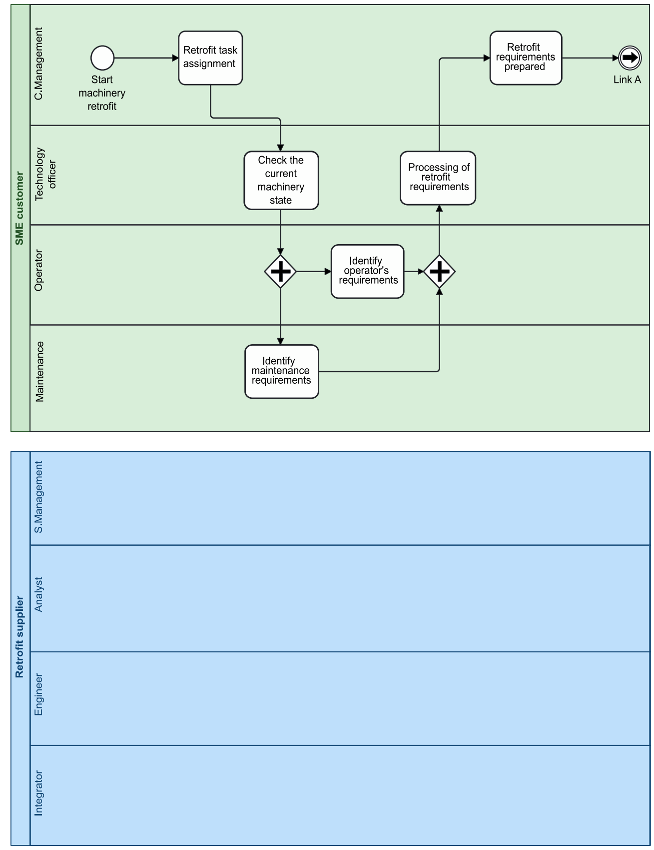

Fig. 6. SME needs identified in Stage 1.

3.1.2. Stage 2 - Requirements assessment and solution design

In the second stage of the process, the specific hardware elements required to provide the required functionality of the system are shown in Fig. 7. These elements are related to the three layers of the automation pyramid. This phase includes the selection and specification of different types of sensors necessary to collect data from the environment, as well as switching, display, control, and HMI elements. Next, actuators that perform physical operations based on control signals and provide mechanical motion to the equipment are defined. A critical part of this phase is the selection of control systems that coordinate and integrate all the elements into a single functional unit. To select and integrate these hardware elements, care must be taken to ensure compatibility, reliability, and expandability to ensure that the system can perform its functions not only in the present but also in the future as technology advances and requirements change.

The verbal description of the BPMN diagram for Stage 2 is as follows: C. Management sends a quotation request for machinery retrofit to the external supplier S. Management in a task (Send retrofit enquiry), where S. Management receives the inquiry and processes it in a task to create a quotation (Quotation task assignment). Next, the Analyst performs the task of analyzing the required activities with a proposal of the type and count of each component (Analyze the requirements and prepare a draft), and the Engineer performs the subprocess (Prepare components list) based on the analysis displayed in Fig. 8.

- Identification of input components: analog, digital, and communication interface devices
- Identification of output components: analog, digital devices, and devices with communication interface
- Identification of control components: the criterion is sufficient computing power, sufficient inputs, outputs, communication interfaces, and other peripherals

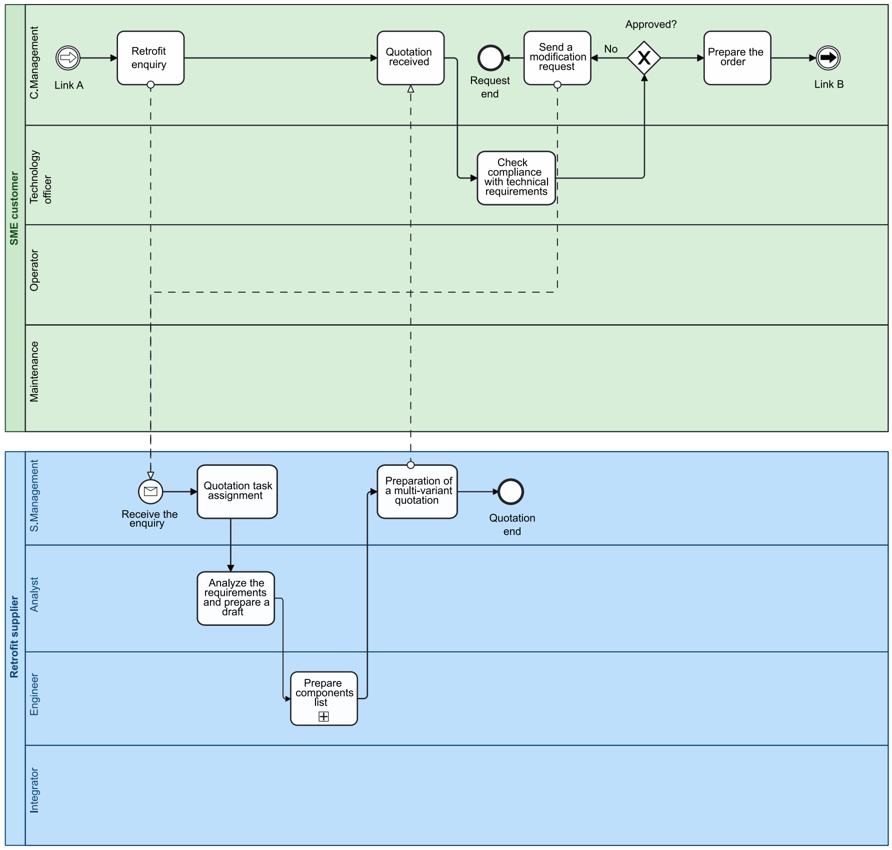

Fig. 7. Requirements assessment and solution design in Stage 2.

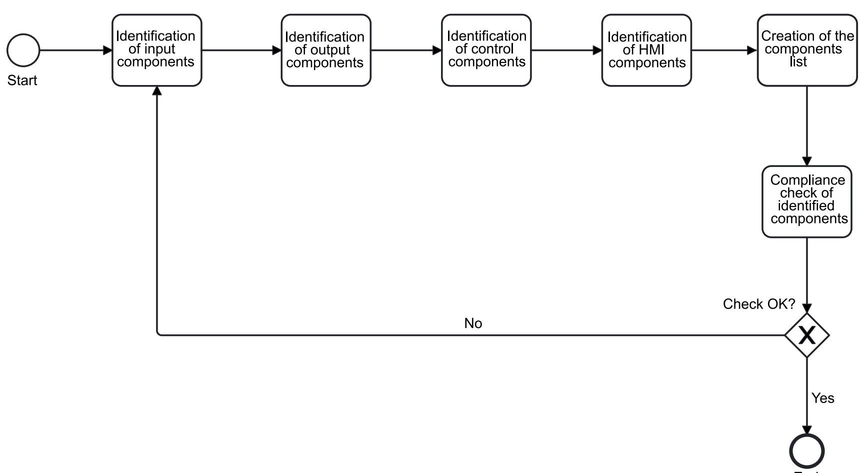

Fig. 8. Identification of components as a subprocess of Stage 2.

- Identification of HMI components: touchscreens, keypads, joysticks, and similar,
- Creation of the components list: based on the identified components
- Compliance check of identified parts: checking the suitability of the selected components, comparing the needs defined for input and output devices and communication and other peripherals with those provided on the control component side

If the check is not OK, the entire process starts again from the beginning (Identification of the input components). If the check is OK, then the subprocess is terminated. Subsequently, S. Management created a quotation (Preparation of a multi-variant quotation) and sent it to C. Management. C. Management receives the quotation (Quotation received) and processes it. Then, the Technology Officer checks the required technical parameters (Check of the required technical parameters). If they do not comply, C. Management requests change to a quotation (Send a modification request). If they comply, C. Management prepares the order (Prepare the order) based on the analysis.

3.1.3. Stage 3 - Implementation of the proposed solution

In the third and final phase, the model output must be implemented, and its functionality and effectiveness must be verified, as shown in Fig. 9. This step involves the installation of new hardware that can be integrated into an existing system.

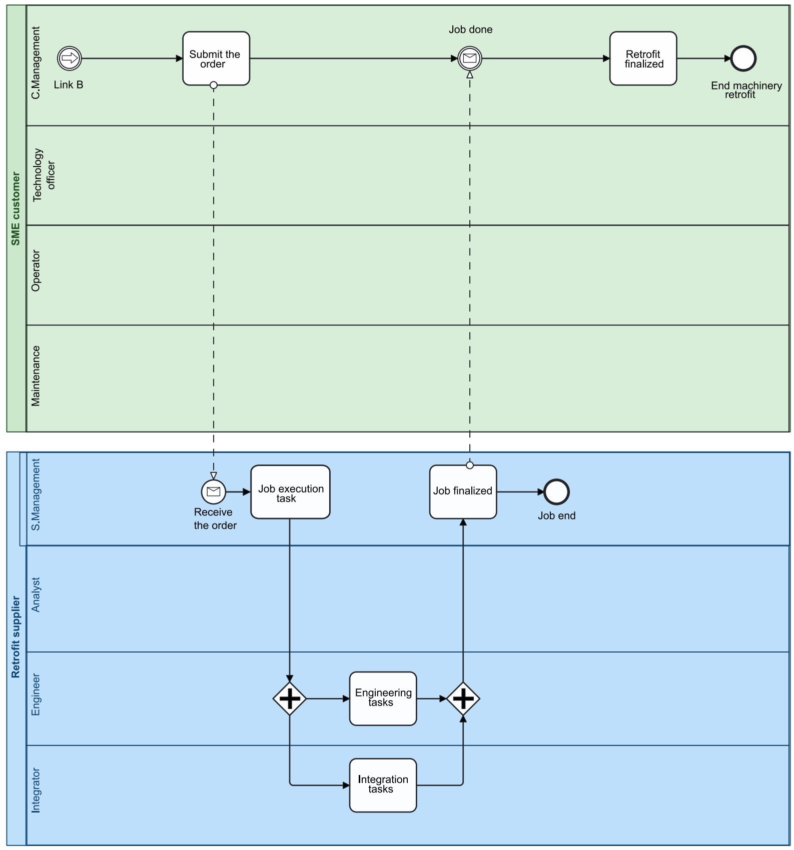

Fig. 9. Implementation of the proposed solution in Stage 3.

Sensors, actuators, and other hardware components must be able to communicate with the control system. This communication is essential for the adequately functioning of the entire system, as it enables real-time data collection and processing, which is critical for controlling and optimizing production processes. This phase is one of the most essential activities in the retrofitting process, as it can reveal incompatibilities between existing sensors, actuators, and new third-party systems. If existing devices cannot communicate with a new platform, they may need to be replaced or modified. It involves additional costs and time but is essential to ensure smooth integration and long-term system functionality. In addition to the technical challenges, implementation must be carefully planned and coordinated to minimize downtime and interruptions to the production processes. Verification involves testing all components and their interactions in different scenarios to ensure that the system performs according to the specifications and is ready for realistic operating conditions. In this way, improvements to existing processes can be achieved, and readiness for future technological innovations and changes in requirements can be ensured.

The verbal description of the BPMN diagram for Stage 3 is as follows: C. Management sends the order (Submit the order), and S. Management receives the order and prepares the job realization (Job execution task). Subsequently, the engineering tasks (Engineering tasks) and integrator tasks (Integration tasks) are executed. When the tasks are completed, S. Management finalizes the job (Job finalized) and sends a message to the C.Management. C. Management receives the completed machine retrofit implementation and closes the retrofit process (Retrofit Finalized).

4. Results and discussion

This section provides an overview of the components and functionality of the AFB production system, and explains the implementation of our proposal in real-world scenario. The AFB (Agro, Food and Beverage is a system integrating multiple stations, advanced control mechanisms, and versatile handling capabilities to streamline the production of both granulate and liquid substances. The AFB production system is a fully automated system comprising 17 stations. Each station was operated using an independent PLC solution based on the Siemens Simatic S7 314C-2 PN/DP series. The system is equipped with electric and pneumatic actuators. This production system was used to fill bottles with granulate or liquid substances. Subsequently, the filled bottles were transported via a conveyor belt on the pallets to the warehouse for dispatch or recycling. The production system facilitates the sorting and preparation of granulate as well as the preparation of liquids. The subsequent section provides a detailed description of the individual operations of the AFB production system, with the block diagram illustrated in Fig. 10. The Granulate Zone: Quality Control, Shaker Conveyor, and Dosage stations were used to sort and process the granulate. The system distinguishes between two types of granulate: yellow (OK) granulate and black (NOK) granulate. In the case of the black granulate, sensors are fitted at the first station to distinguish the color, and the black granulate is then blown out of the system and placed in a collection container. The sorted granulate was then processed and prepared in a plastic container for bottling at a bottling station. The Liquid Zone — Filtration, Mixing, Reactor, and Quality Probe stations are used to process the liquids before bottling. During the processing, the liquid undergoes filtering, mixing (where the system allows different types of liquids to be mixed), and heating or cooling to the required temperature. The processed liquid can then be sampled for quality testing. The processed liquid was pumped into a plastic container at a bottling station. The Lids Zone — Distribution, Buffer, and Handling stations were used to move the bottle caps to the Bottling Station where the bottles were capped. The inlet for the lids was a buffer filled by a robot from the Robot Recycling station. Two pneumatic arms and a conveyor belt were used to move the lids.

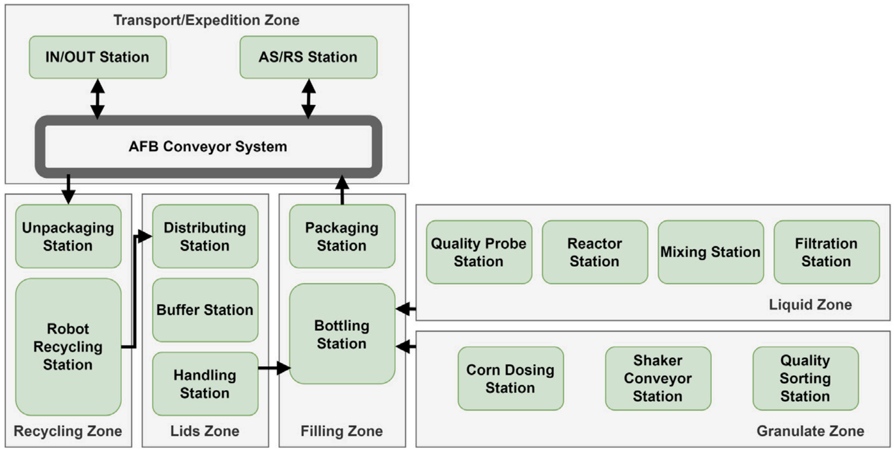

Fig. 10. AFB production system block diagram.

The Filling Zone — the Bottling station was used to fill the bottles with granulate and liquid. At the entrance to this station is located a conveyor belt with bottles. The bottles were separated in two rows and are alternately dropped onto a conveyor belt. The bottles were divided into granulate bottles and liquid bottles. The bottles were then fed into a rotating carousel and identified by an RFID reader before entry. The bottles were then filled with either granulate or liquid, and the system could fill the bottles to either 0, 50 or 100 percent according to the order placed. A cap was brought in from the Handling Station to the filled bottles further in the process, and the bottles were sealed using a rotating pneumatic arm. The filled and capped bottle was moved to an RFID reader, where the information was written on the bottle. The bottle thus prepared proceeds to the Packaging station, which transports and loads the bottles onto the pallets to the Conveyor System station. The Transport/Expedition Zone — the Conveyor System station serves as a conveyor belt for transporting filled and sealed bottles. The pallets were transported between four stations: Packaging, AS/RS, IN/OUT and Unpackaging. The AS/RS station serves as an intermediate storage station for filled pallets. It can store up to 12 pallets using a three-axis manipulator. The manipulator can remove the pallets from the warehouse and place them on an empty trolley at the Conveyor System station. The IN/OUT station serves as a dispatch station and is an entry for empty pallets. This station also operates with a three-axis manipulator that can either take pallets off the conveyor belt and send them to dispatch, or load empty pallets back onto the conveyor belt. The Recycling Zone — Unpackaging and Robot Recycling stations are used for recycling filled bottles. If the bottles are destined for recycling, the Unpackaging station takes the bottles off the pallet from the conveyor system. An RFID reader then identified the bottles to detect the material in the bottle. The Robot Recycling station unscrews the bottle identified with the help of the robot and returns the cap to the bin at the Distributing station. After handing over the cap, the robot takes the appropriate attachment to extract the granulate or liquid, and extracts the material from the bottle into the proper container. The empty bottle was moved to the Bottling station inlet, where an RFID reader was used to record the emptying of the contents, and the bottle was moved to the appropriate bin for refilling.

Based on operational experience, bottlenecks in the system have been identified, specifically at the Bottling and Robot Recycling stations and locations where occasional product jams occur. One of these bottleneck locations was the inlet at Bottling station. As the station is also a bottleneck in the production system, it can be identified as a critical point in the production process. When entering the rotary carousel, bottles are jammed between the conveyor belt and carousel, causing the entire production to stop until the fault is rectified. The fault can only be eliminated by manually pulling out the jammed bottle that the line operator must perform. Another identified location where occasional product jams occur is the Conveyor System station. At this location, trolley jamming occurs on the conveyor belt near the Unpackaging station, but is not as critical as at the Bottling station. In most cases, the jammed cart moves further down the belt when the incoming cart pushes it. However, operator intervention is required if the trolley is stuck alone on the belt.

Fig. 11. Bottling station critical point.

4.1. Model application

Based on the identified problems, we decided to retrofit the equipment according to the proposed solution to eliminate them. This problem was solved by two teams of two people, with each person performing two roles. One team represented the SME Customer, and the other represented the Retrofit Supplier. In SME Customer, the first person performed C. Management and Technology officer roles, and the second person performed Employee and Maintenance roles. In the Retrofit Supplier, the first person performed S. Management and Analyst roles, and the second person performed Engineer and Integrator roles. This section describes the retrofit performed at the Bottling Station, because it is a critical location in the AFB production system. Fig. 11 shows the Bottling station, where the black granulate bottle arrives at a critical location where the bottles are jammed.

The entire process started with the first phase by C.Manager assigning the task to the Technology Officer as a requirement to arrange the retrofit of the production line. The Technology Officer assigned

Table 7
List of components used in AFB upgrade.

Category	Component	Quantity
Input component	Accelerometer ICM20948	2
Input component	Microphone SPH0645	2
Input component	Raspberry Pi 4	2
Output component	Accelerometer data	2
Output component	Audio data	2
HMI component	–	–

Employee tasks to identify the employee requirements and Maintenance to identify the current machinery status. Maintenance identified the machinery state, and the output was the bottlenecks and critical points of the system and the errors that occurred on the line, as described above. The output also includes the current state of the line, with a focus on Industry 4.0. The AFB system was free of Industry 4.0 elements before the retrofit and process data were collected from the PLC. At the same time, the output from the Employee was the requirement that the production system be able to alert the operator when a critical error, such as a production jam at the Bottling station, occurs, thus speeding up the resolution of the problem. The Technology Officer summarized the outputs from both Employee and Maintenance and escalated to C.Management to send a quotation request to S. Management of Retrofit Supplier. Simultaneously, the process proceeds to the second phase.

S. Management accepted the inquiry and created a role for the Analyst. The Analyst analyzed the requirements and drafted a solution based on sensing vibrations and sounds at the bottlenecks and critical points accompanying line jams. However, the new sensors cannot be connected to the existing PLC architecture because this architecture is a closed system to which we do not have full access, and thus, cannot modify the program. Based on this insight, the Analyst chose to use a circuit with microcontrollers as a solution. The draft prepared in this manner was forwarded to the Engineer. The Engineer select components based on the draft according to the proposed subprocess. A Raspberry Pi 4 microcontroller was used as the Control Component. For the Input Components, Accelerometer Sparkfun ICM20948 and Microphone SPH0645 were chosen. The Output Components, in this case, are accelerometer and audio data. The HMI in the production system case does not need to be incorporated into the solution. The selected components are presented in Table 7.

Based on the draft solution and the selected components, S. Management prepared a retrofit solution quotation and sent it to C. Management. On the SME side, the Customer Technology Officer evaluated the technical parameters of the proposed solution as satisfactory and C. Management approved the solution. At the same time, a retrofit order was sent to S. Management of the Retrofit Supplier, bringing the process to its final stage. S. Management assigns tasks to the Engineer and Integrator. The Raspberry Pi 4 microcontrollers were mounted on the production system close to the locations where the errors mentioned above occurred. Sensors were attached to the control boards, where one accelerometer and one microphone were used to pick up specific sounds and vibrations in the production system. Fig. 12 shows the fitted Raspberry Pi 4 microcontroller and Fig. 13 shows a closer view of the fitted and connected SPH0645 microphone.

Based on these sensors, the microcontrollers can provide data that can be evaluated and alert the line operator in the case of an AFB error. After the processes from the Engineer and Integrator are completed, the retrofit process is finalized and completed by C. Management. As mentioned above, before the retrofit started, the operator had to personally detect that the production line was stuck and manually correct the error. This applies to both Bottling and Conveyor System stations. Approximately every 30 min, the operator refills the granulate buffer and tops the liquid. If the problem occurred immediately after refilling the raw materials, it could have caused a production downtime

Fig. 12. Raspberry Pi4 position.

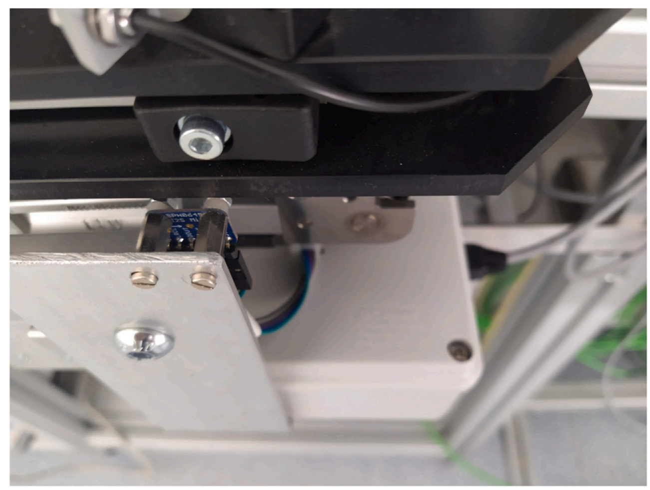

Fig. 13. SPH0645 microphone position.

of up to 30 min because the operator left to perform other tasks. After the retrofit, the line downtime was reduced to a maximum of 3 min, with the operator receiving notifications in the MES system as soon as an error occurred. As this proposal only deals with the hardware solution of the retrofit, the software solution is not part of it.

This successful implementation demonstrates the potential retrofitting existing production systems with advanced technologies to meet the evolving demands of modern manufacturing. The AFB system now stands as an example of how traditional production lines can be effectively upgraded to incorporate smart manufacturing principles, enhancing both efficiency and competitiveness in the agriculture, food, and beverage sectors.

The retrofitting solution involved the integration of Raspberry Pi 4 microcontrollers equipped with accelerometers and microphones. These components were strategically placed at critical points in the system to monitor the vibrations and sounds associated with the jams. The inability to modify the existing PLC architecture necessitates this independent microcontroller-based approach. The retrofit significantly improved the system performance by enabling real-time detection of errors and reducing the downtime from up to 30 min to a maximum of 3 min. Notifications to operators via the MES system ensured rapid intervention, thus enhancing overall system efficiency. Hardware-focused retrofit demonstrates the potential for targeted technological upgrades to address specific operational challenges within automated systems.

This solution underscores the importance of adaptability in industrial automation by leveraging modern sensing technologies and bypassing the limitations of legacy architectures. Future work may extend to software integration, enhanced data analysis, and predictive maintenance capabilities to further optimize the AFB production system.

5. Conclusion

This manuscript presents a comprehensive analysis of the challenges faced by SMEs in the European Union as they navigate the transition towards Industry 4.0. This effectively highlights the economic pressures on SMEs and the importance of modernizing their operations to remain competitive. The statistical data provided a strong foundation for understanding the significance of SMEs within Industry 4.0, alongside the EU economy. The retrofit of the AFB production system based on the proposed BPMN model addressed critical inefficiencies and bottlenecks at key points in the production process, particularly at the Bottling and Conveyor System stations. By implementing modern hardware solutions, such as Raspberry Pi microcontrollers equipped with accelerometers and microphones, the system was able to detect disruptions in real-time, providing timely alerts to operators and significantly reducing downtime. This upgrade not only enhanced operational efficiency but also demonstrated the effectiveness of integrating new technologies into legacy systems. The use of microcontrollers as standalone solutions bypassed the limitations posed by the closed PLC architecture, showcasing the adaptability of modern technology to existing infrastructure. The successful reduction in downtime from up to 30 min to just 3 min illustrates the value of targeted hardware interventions in minimizing production losses and maintaining consistent output. This article highlights the importance of a systematic approach to industrial retrofitting, including the thorough identification of bottlenecks, careful selection of appropriate technologies, and collaborative efforts between system stakeholders. Although the hardware implementation achieved significant improvements, future efforts could focus on integrating software solutions for predictive maintenance, enhanced data analytics, and further automation of error resolution. AFB retrofit serves as a model for addressing similar challenges in other automated production environments, emphasizing the balance between technological innovation and practical implementation. Building upon our previous work Barton et al. (2022), we will investigate the implementation of AI techniques into industrial processes specifically tailored for SMEs, reflecting the current state of AI capabilities and their practical application in manufacturing environments. This includes developing machine learning algorithms for predictive maintenance, standardized communication protocols between retrofitted components, and scalable data management architectures suitable for SME environments.

CRediT authorship contribution statement

Gabriel Gašpar: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Roman Budjač:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Maroš Valášek:** Writing – original draft, Methodology, Investigation. **Martin Bartoň:** Writing – original draft, Validation, Investigation. **Tomáš Meravý:** Visualization, Validation, Software, Methodology. **Maximilián Strémy:** Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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